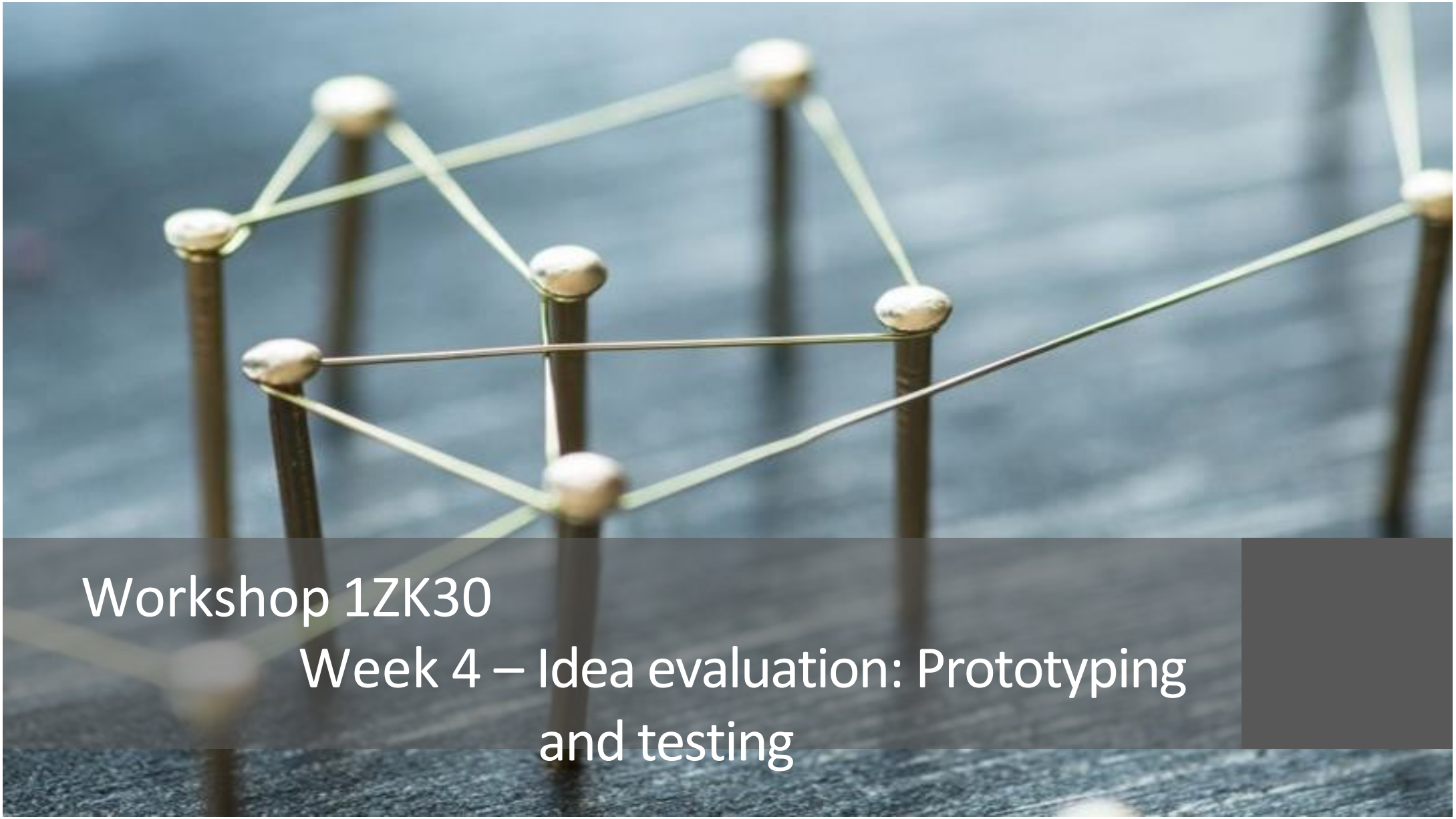
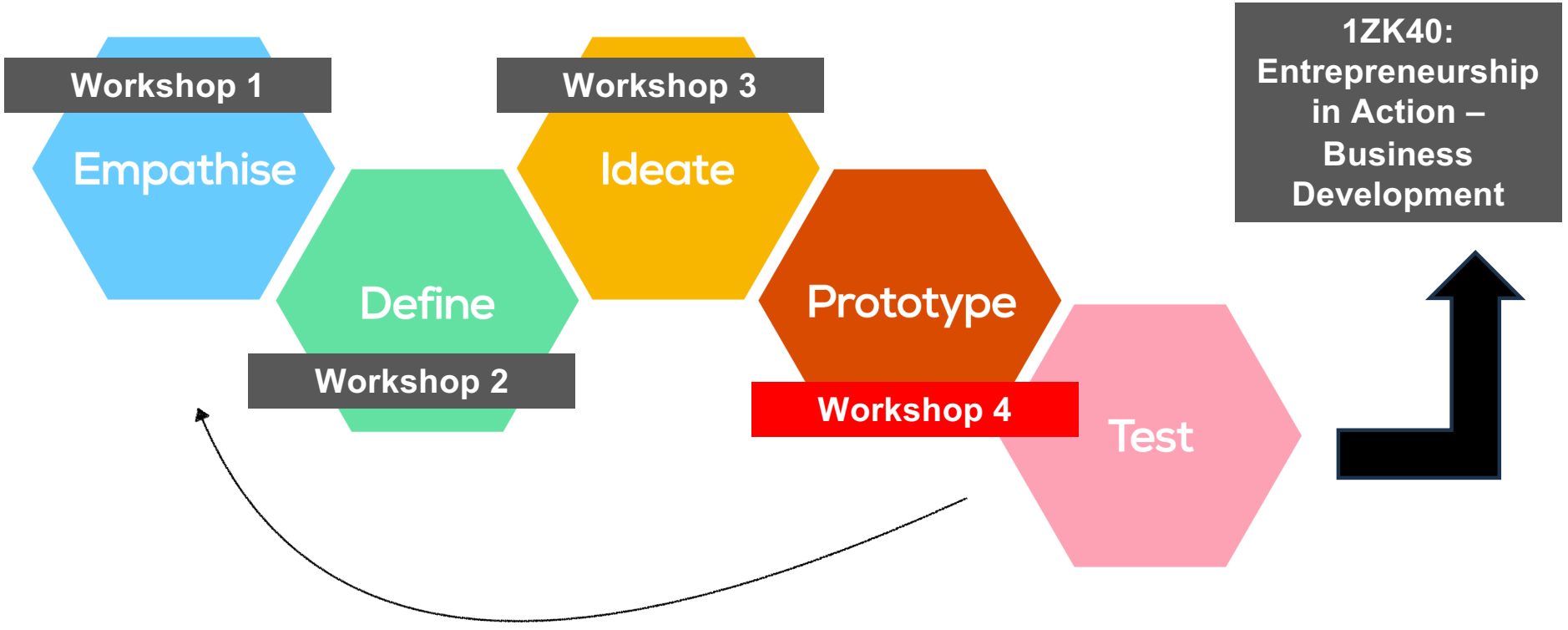


Workshop 1ZK30

Week 4 – Idea evaluation: Prototyping and testing

DESIGN THINKING



RECAP

Workshop 1 – Team & Problem Exploration

- You formed teams and you defined your strengths, roles & rules
- You explored neighborhoods (data detective, journalist, anthropologist) and collected first insights

Workshop 2 – Problem Identification

- You analyzed interviews, observations, and archival data
- You developed personas and crafted problem statements based on your data

Workshop 3 – Ideation

- You applied creativity techniques to generate a wide variety of ideas
- You clustered, prioritized, and selected 3–5 promising ideas
- You developed your first Idea Canvas for the most promising concept

Together, these outputs define WHO you design for, WHAT problem matters, and HOW you might solve it.

➢ **This is the foundation for Workshop 4:** Idea Evaluation – prototyping and testing your ideas to assess desirability, feasibility, and viability.

DESIGN THINKING

Prototype

A prototype is a simple, inexpensive, and quick way to develop ideas so that you can interact with and experience them.

- **Start building:** Build a low-resolution sketch. It can be a physical object or a digital click dummy (quick & dirty).
- **Storyboard:** Design a scenario in which people can experience the solution in a physical environment.



DESIGN THINKING

Test

Ask for feedback on your prototype. Learn about your users, refocus your attention, and refine your prototype.

- **Presentation:** Let people use your prototype. Put it in their hands and listen to them.
- **Create experiences:** Let people talk about their experiences and feelings.



OBJECTIVES OF WORKSHOP 4

1. Understand challenges of evaluating innovative ideas (bias, uncertainty, feasibility).

2. Prototype one idea quickly (e.g., sketch, mock-up, storyboard).

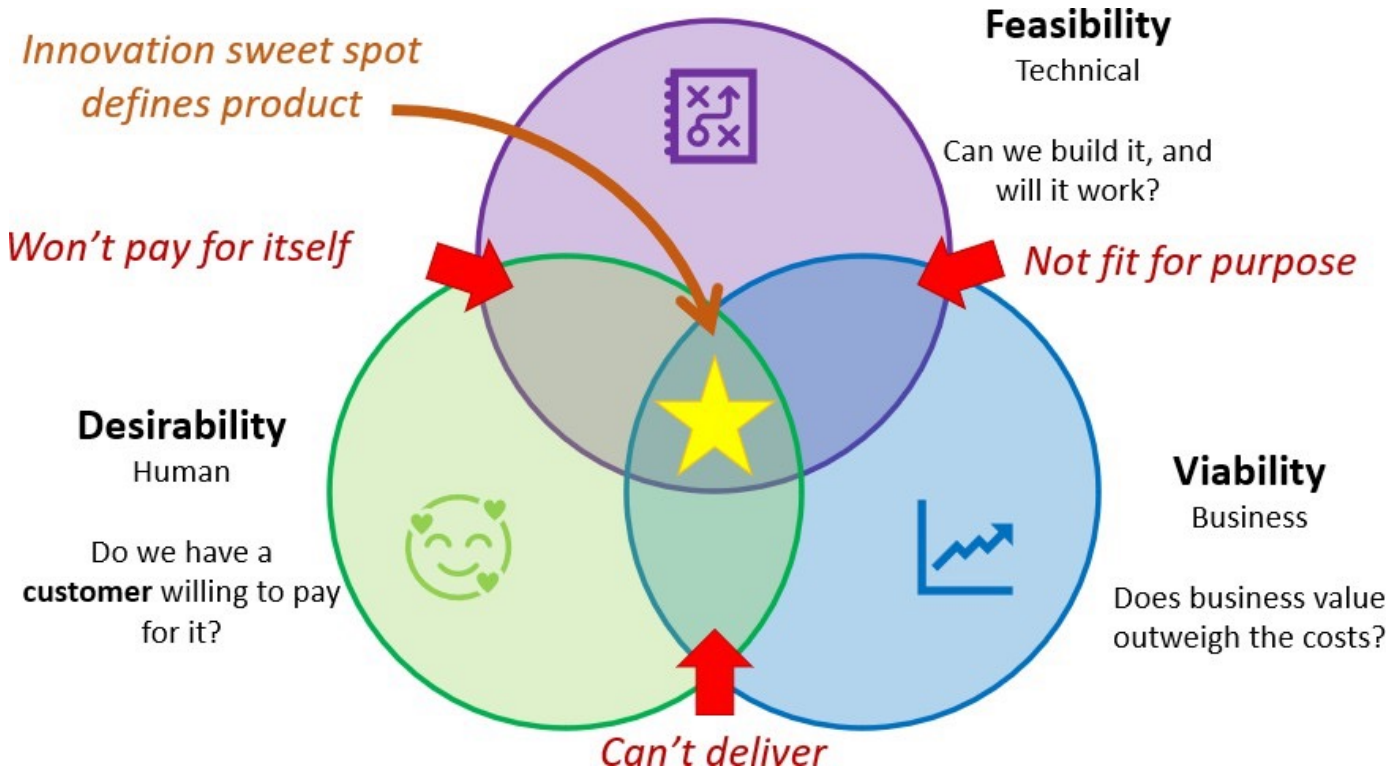
3. Learn and practice concept testing to assess desirability of solutions.

4. Learn and apply the Market Opportunity Navigator to assess feasibility and viability.

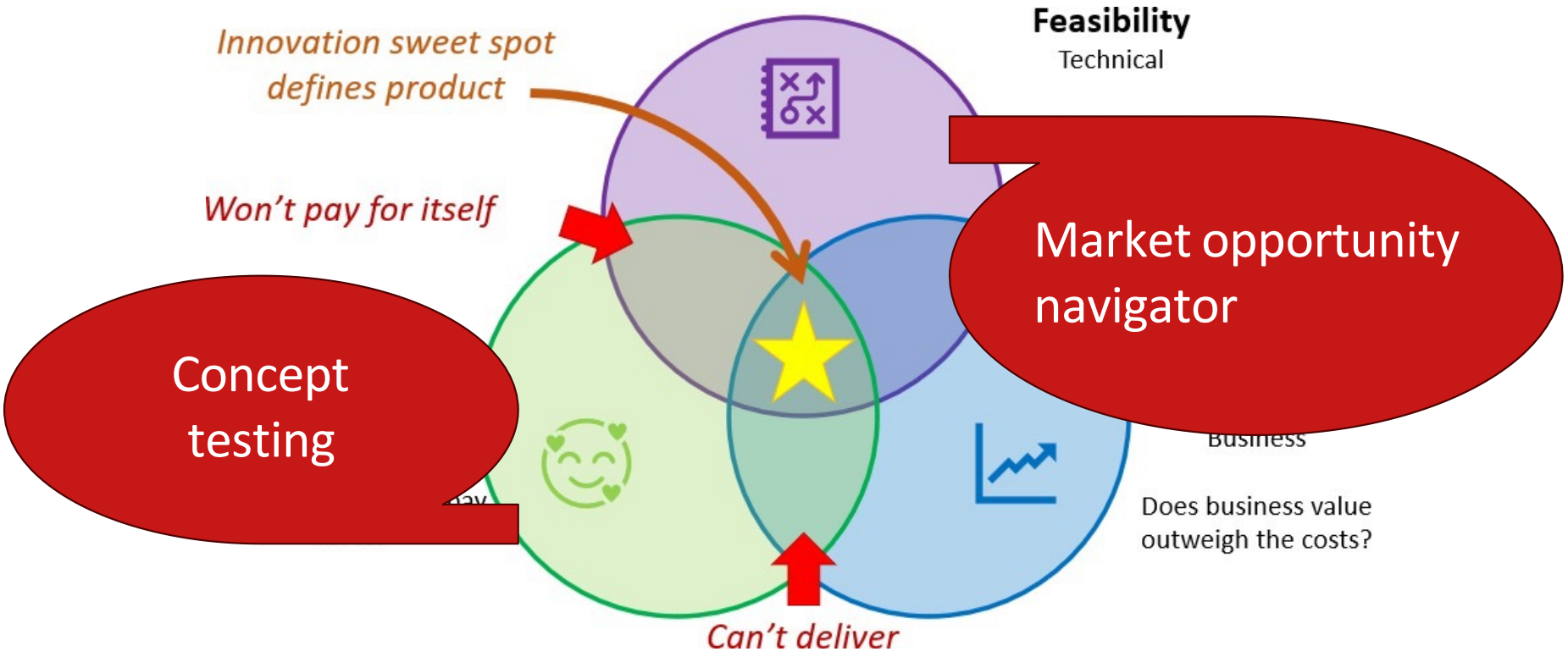
AGENDA

1. Introduction: Idea evaluation and the role of experiments
2. Prototyping
3. Break
4. Concept Testing (Assessing Desirability)
5. Market Opportunity Navigator (Assessing Feasability & Viability)
6. Wrap-up & Deliverables for Assignment 2

EVALUATING IDEAS



EVALUATING IDEAS



EVALUATING IDEAS

Even expert assessments are often short-sighted and flawed:

Ferdinand Foch, French Military Strategist, 1911

"Aircrafts are interesting toys, but they have no military value."



Harry Warner, Founder of Warner Brothers, 1927

"Who the hell wants to hear actors talk?"



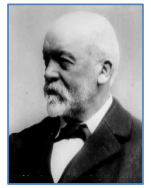
Thomas Watson, IBM's CEO, 1943

"I think there is a world market for about five computers."



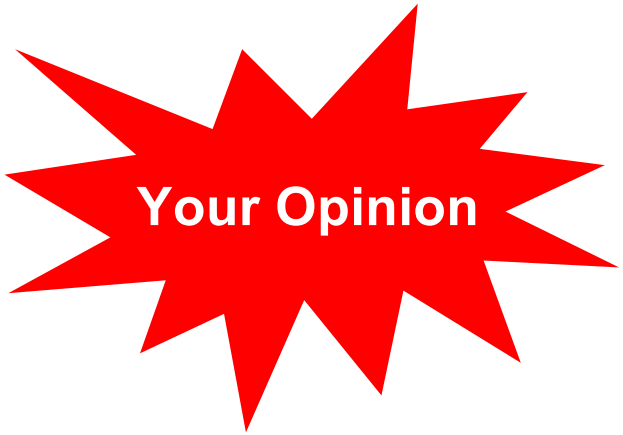
Gottlieb Daimler, Founder of automobiles, 1901

"Worldwide demand for automobiles will never exceed one million - simply because of the lack of available chauffeurs."



EVALUATING IDEAS

Why do you think it is so difficult to evaluate new ideas?



EVALUATING IDEAS

Challenges in assessing new ideas:

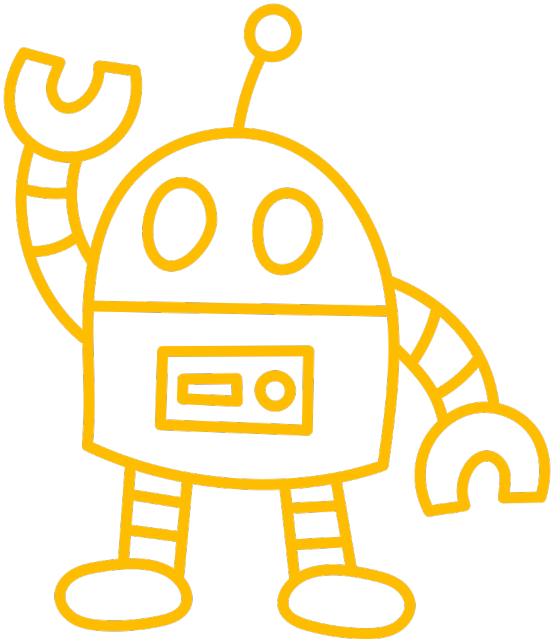
Uncertainty and ambiguity

- Long time horizon from initial idea to market launch
- Possibility of unexpected combinations (e.g., the invention of Post-It Notes was an "accident" as the goal was to create a very strong adhesive)
- Probabilities of success often cannot be reliably estimated (many variables)

Local evaluation bias

- Evaluators are guided by the past, and the evaluation criteria reflect past success factors
- Unwillingness to adopt ideas that were invented elsewhere (not invented here syndrome)
- Unwillingness to accept (disruptive) ideas that threaten the current business model (cannibalization, e.g., the Blockbuster video store vs. streaming providers)

EVALUATING IDEAS



Dealing with uncertainty through efficient experimentation and prototyping:

"Maximize the rate of learning by minimizing the time to try ideas“

by Tom Chi at TEDxKyoto

EVALUATING IDEAS



Understanding the benefits of experiments

Definition and relevance

- Trial and error process in which each attempt provides new insights into a problem
- The result of each experiment is an important learning input
- Essential tool for problem solving in situations of uncertainty and imperfect information
- It has been found that R&D teams spend more than 70% of their time on experiments

EVALUATING IDEAS

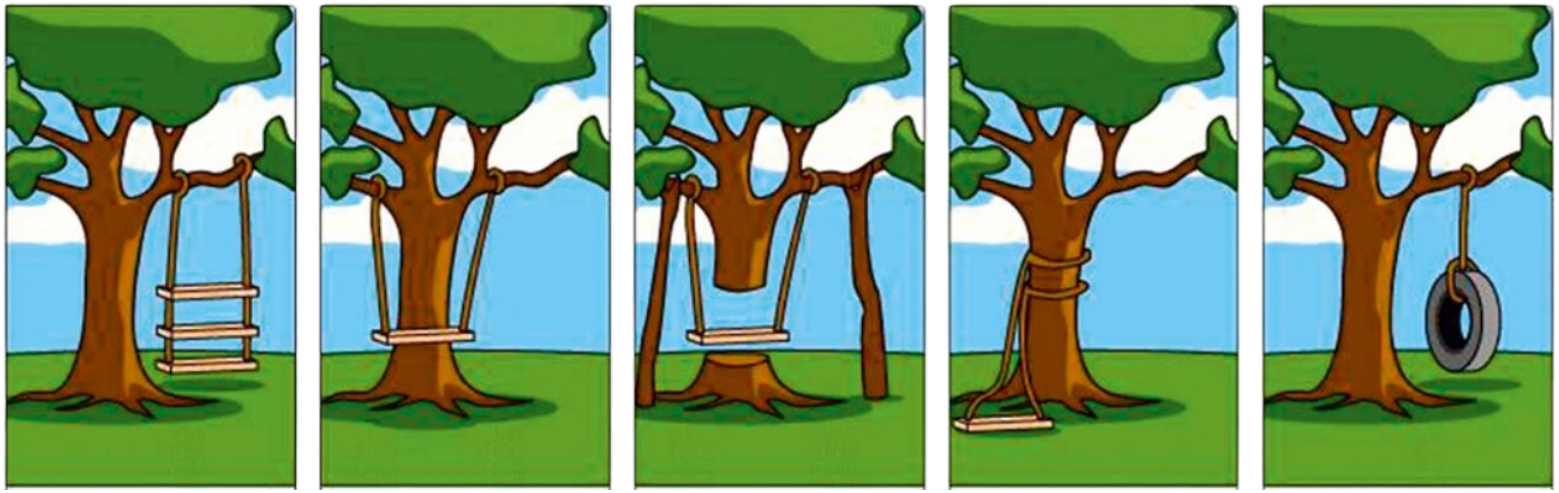


Understanding the benefits of experiments

Small losses strategy

- The costs of experiments and failed attempts are accepted as a necessary investment in learning and problem solving.
- Small losses strategy as a means of minimizing the risk of a major failure
- Cycles of small-scale testing, refinement, retesting, and scaling

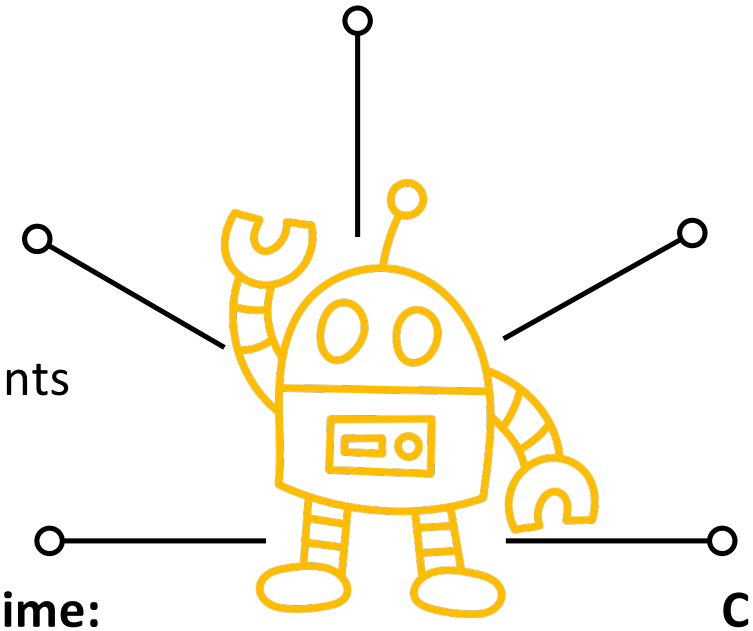
EVALUATING IDEAS



How the project was initially described	How the project leader understood the project	How the implement-ation was planned	What was programmed/ coded	What the customer actually needed

IDEA EVALUATION | IMPLICATIONS OF EXPERIMENTS

Error correction: Fix costly errors early on



Test phases:

Test different variants

Cost savings:

Save costs in production

Development time:

Shortening of development times

Check of product potential:

Obtain early feedback & convince stakeholders

PROTOTYPING

*“We define **prototype** as any representation of a design idea, regardless of medium”* *(Houde & Hill, 1997)*

In other words, a prototype doesn't have to be a physical object or a fully functional product. It could be:

- A sketch
- A digital mockup
- A (clickable) wireframe
- A paper, wood or foam model
- A storyboard
- Etc.

"Make sure you are building the right 'it' before you build 'it’”
by Alberto Savoia (2011)

Houde, S., & Hill, C. (1997). What do prototypes prototype?. In *Handbook of human-computer interaction* (pp. 367-381). North-Holland.

PROTOTYPING

- **Role:** product/service function into users' lives.
- **Look and feel:** sensory experience of user.
- **Implementation:** technical functioning, how it actually works.

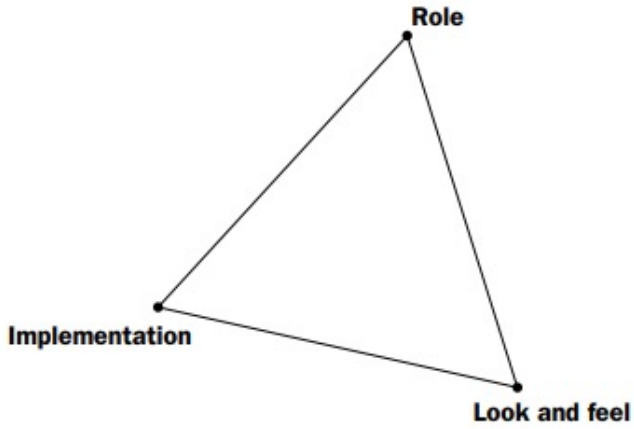


Figure 1. A model of what prototypes prototype.

Lauff, C. A., Kotys-Schwartz, D., & Rentschler, M. E. (2018). What is a Prototype? What are the Roles of Prototypes in Companies?. *Journal of Mechanical Design*, *140*(6), 061102.

PROTOTYPING

"Prototypes must first be tools for communication and learning, sometimes sequentially or in parallel, before they can better inform decision-making."

(Lauff, Kotys-Schwartz & Rentschler*, 2018)*

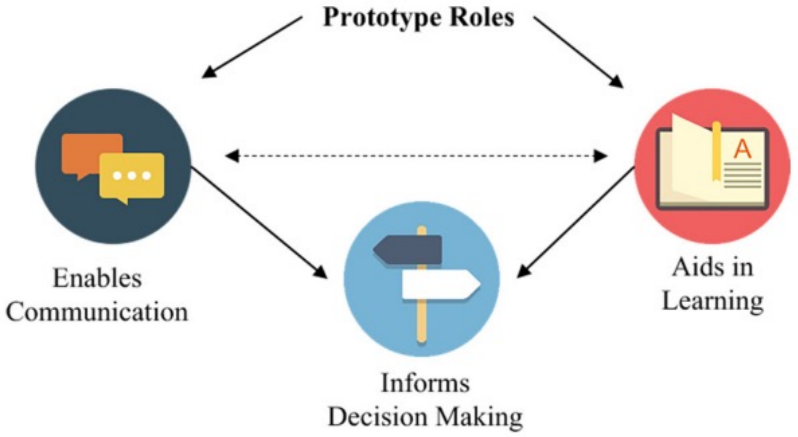


Fig. 5 Three emergent roles of prototypes: communication and learning to support decision-making

Lauff, C. A., Kotys-Schwartz, D., & Rentschler, M. E. (2018). What is a Prototype? What are the Roles of Prototypes in Companies?. *Journal of Mechanical Design*, *140*(6), 061102.

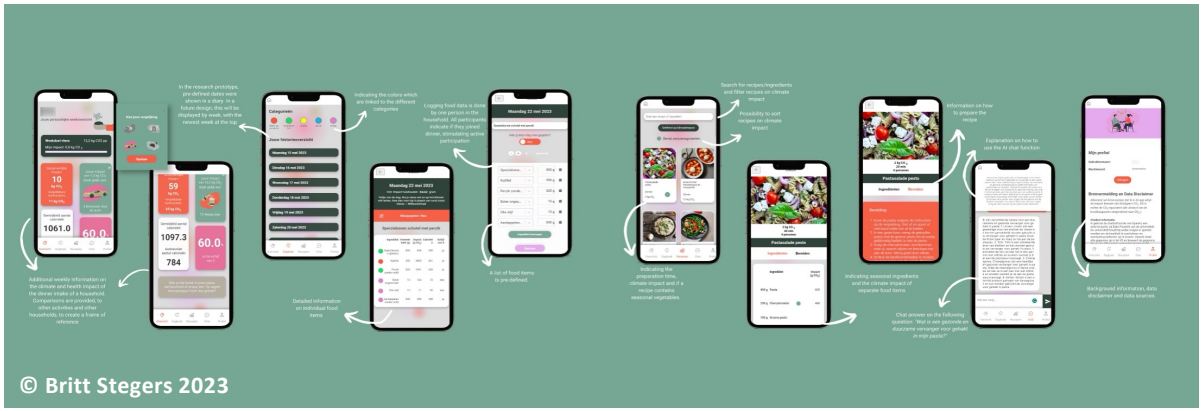
PROTOTYPING EXAMPLES

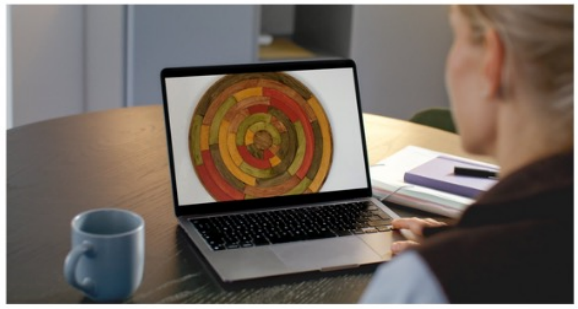
Digital Mockup



Canva

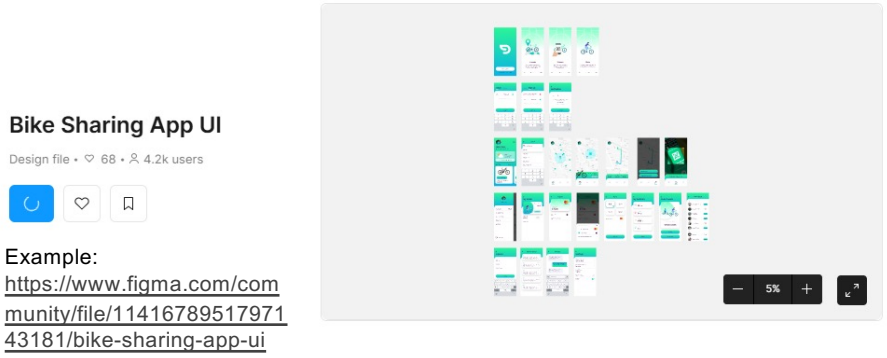
Home - Canva



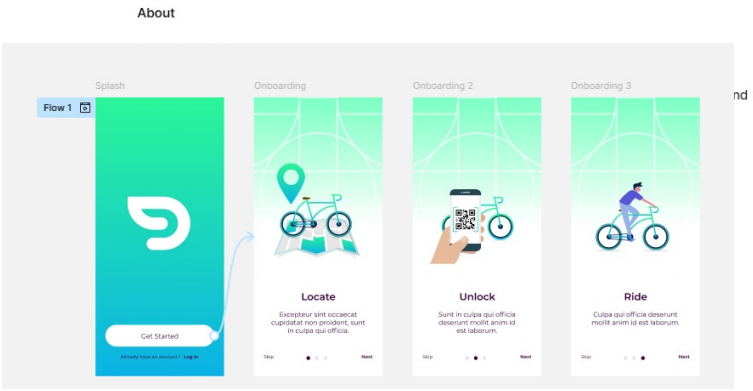


PROTOTYPING EXAMPLES

A (clickable) wireframe

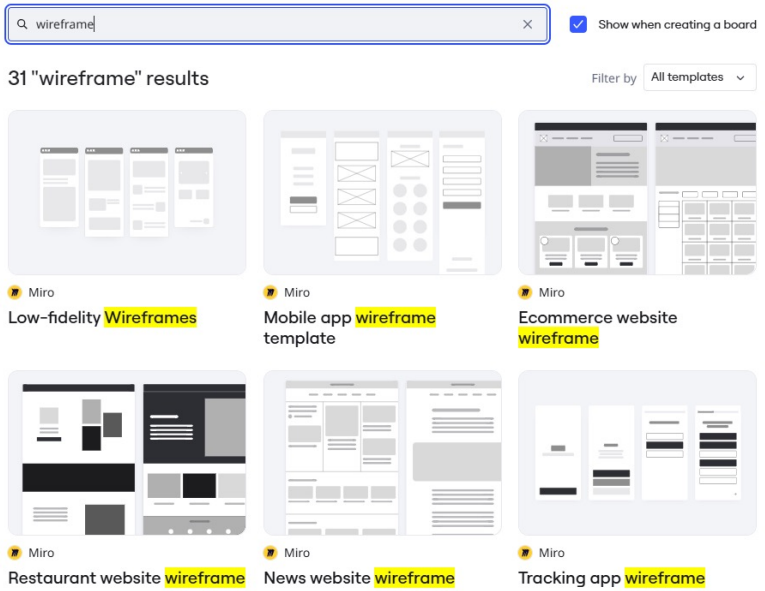


Example:
https://www.figma.com/community/file/1141678951797143181/bike-sharing-app-ui



Figma

http://figma.com/



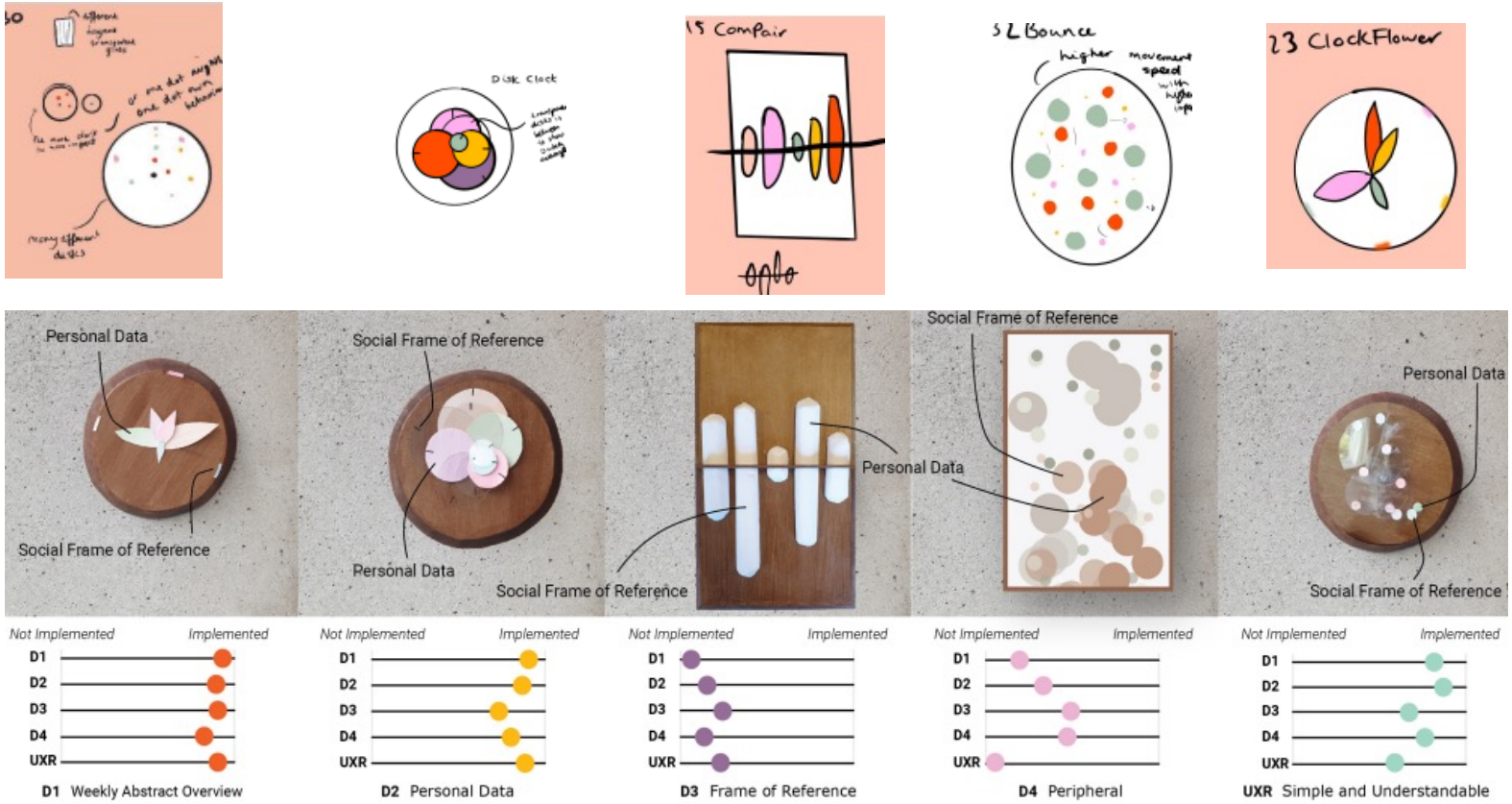
miro http://miro.com/

PROTOTYPING EXAMPLES

A paper, wood or foam model

From sketch to prototypes

Analyzing value for users through design principles



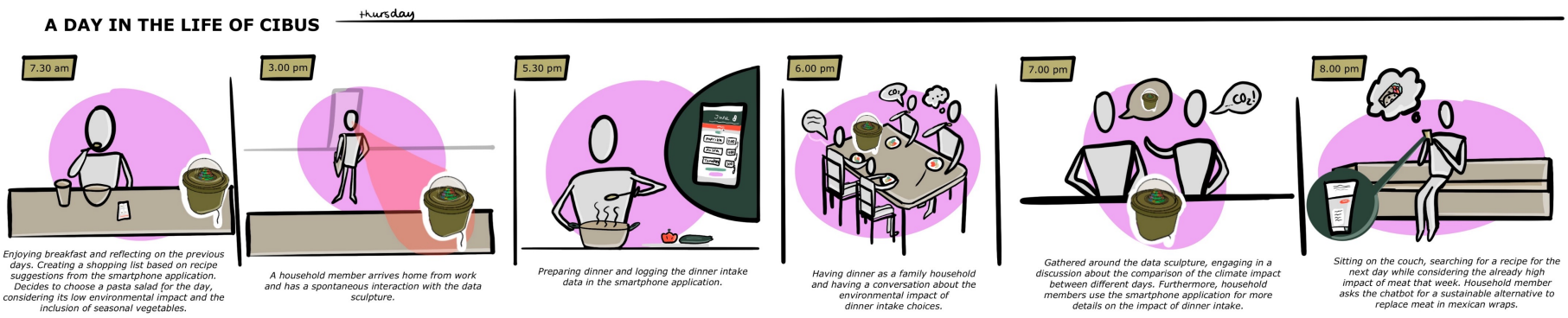
Stegers, B., Sauvé, K., & Houben, S. (2022, June). Ecorbis: A data sculpture of environmental behavior in the home context. In *Proceedings of the 2022 ACM Designing Interactive Systems Conference* (pp. 1669-1683).

PROTOTYPING EXAMPLES

A storyboard



© Britt Stegers 2022



© Britt Stegers 2023

EXAMPLE OF PROTOTYPING PROCESS



Start with

doodles



Expand with

Sketching



Finalizing with

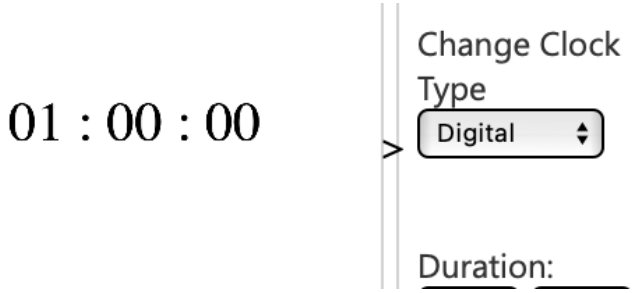
wireframes

Low **Fidelity** High

LET'S PROTOTYPE



60 minutes



Create a sketch, mock-up (or physical prototype) for one of your 3 selected ideas:

- Start with sketching! Afterwards you could start with a digital or physical prototype
- Potential supportive tools: e.g., Miro, Canva, Figma, Tally
- Physical prototyping: e.g., cardboard, scissors, paper, pencils etc.
- How is the prototype used in the user's life?
- **Criteria: Is the solution easy to understand? Does it address the needs? Is it attractive/desirable?**

BREAK



IDEA EVALUATION

Testing desirability









CONCEPT TESTING



What: Leverage concept sketches to judge how well a solution direction attracts your potential users



Objective: identify whether a chosen solution appeals to potential customers / users and their needs



Who: participants who are able to verbalize their opinion about whether the concept meets their needs

CONCEPT TESTING: HOW?



Create a mockup or sketch for the 3 ideas resulting from the ideation workshop



Recruit a minimum of 5 participants that a (for each prototype) that are willing to evaluate these solutions (participants should be selected based on your target audience/potential customer)



Qualitative: Present the concept briefly (e.g., 1 minute), subsequently hide the sketch or mock up and ask them to describe what they remembered about the concept and if it held any appeal. Afterwards, let them ask questions and use/experience the prototype. Finally, generate feedback (attractive? needs potentially addressed?). Repeat this for every of the 3 retained ideas



Quantitative: After talking to a potential user and testing the concept qualitatively, let the same user fill in the short version of the "User Experience Questionnaire (UEQ-S)" (https://www.ueq-online.org), which is available in 30 languages, including Dutch. You can directly download the questionnaire here (https://www.ueq-online.org/Material/UEQS_Items.pdf) and a data analysis tool here (https://www.ueq-online.org/Material/Short_UEQ_Data_Analysis_Tool.xlsx).



Sit down as a team and go through the various concepts presented to the participants and rank and evaluate the ideas based on user desirability (i.e., qualitative and quantitative feedback).

CONCEPT TESTING (Quantitative)

Short Version of the User Experience Questionnaire (UEQ-S):

English version

obstructive	o o o o o o o	supportive
complicated	o o o o o o o	easy
inefficient	o o o o o o o	efficient
confusing	o o o o o o o	clear
boring	o o o o o o o	exciting
not interesting	o o o o o o o	interesting
conventional	o o o o o o o	inventive
usual	o o o o o o o	leading edge

Instructions and more information about the UEQ: https://www.ueq-online.org

Download link for the questionnaire: https://www.ueq-online.org/Material/UEQS_Items.pdf

Download link for the data analysis tool: https://www.ueq-online.org/Material/Short_UEQ_Data_Analysis_Tool.xlsx

CONCEPT TESTING IN ACTION



20 minutes

20 : 00 : 00

Change Clock Type

Digital

Duration:

Present your prototype to another team (e.g., team 1 with team 2, team 11 with team 12 etc.), and generate feedback:

Qualitatiive:

- Present the concept briefly, subsequently hide the sketch or mock up and ask them to describe what they remembered about the concept and if it held any appeal.
- Afterwards let them ask questions and use/experience the prototype.
- Finally generate feedback: e.g., is it easy to understand? Does it address the needs? Is it attractive/desirable?

Quantitative:

- Let the same users fill in the User Experience Questionnaire (short)

Based on this test/feedback: What works? What does not work? What should be adapted?

OPPORTUNITY EVALUATION

Testing feasibility and viability

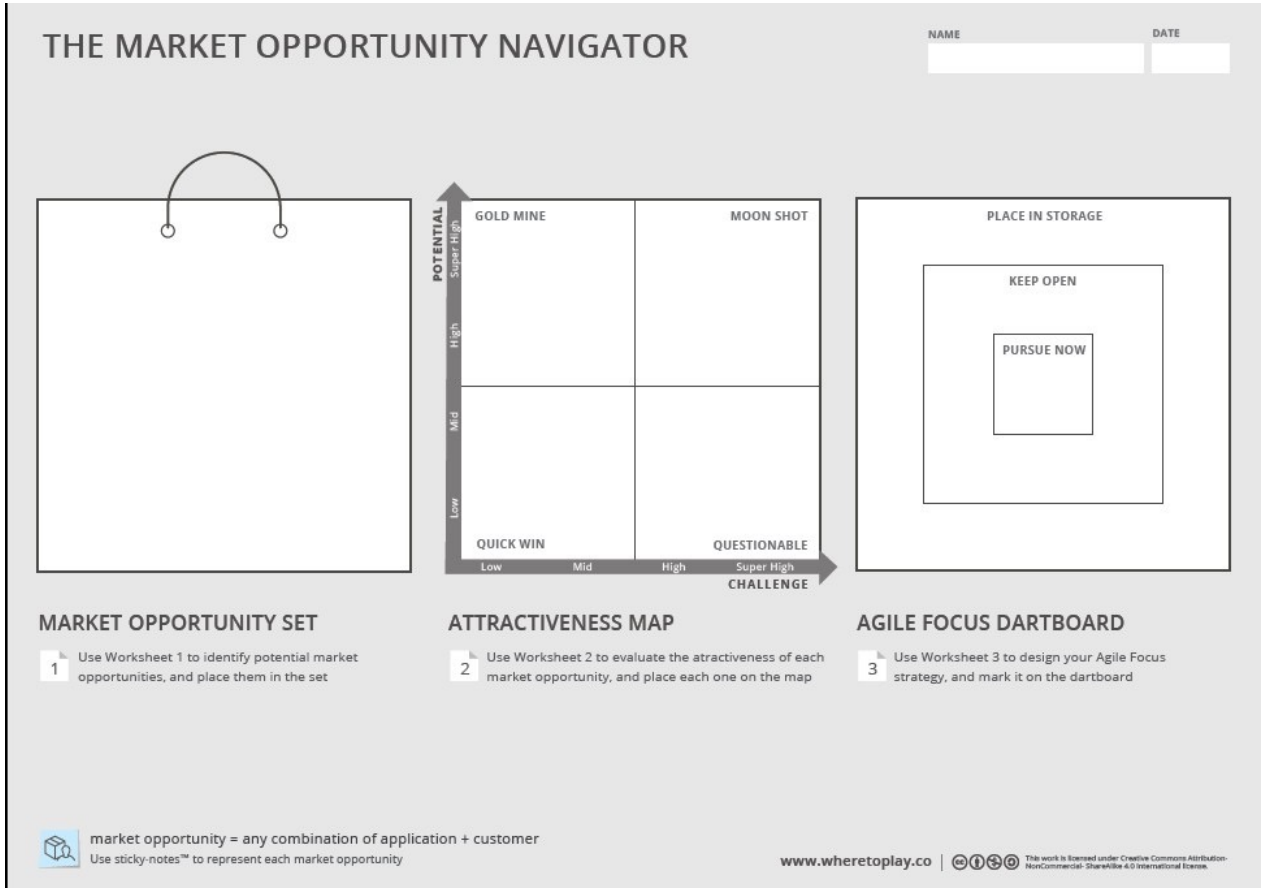


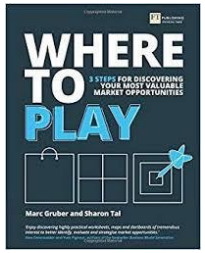
MARKET OPPORTUNITY NAVIGATOR

Approach for discovering the most valuable market opportunities

Targeted towards technology-driven solutions

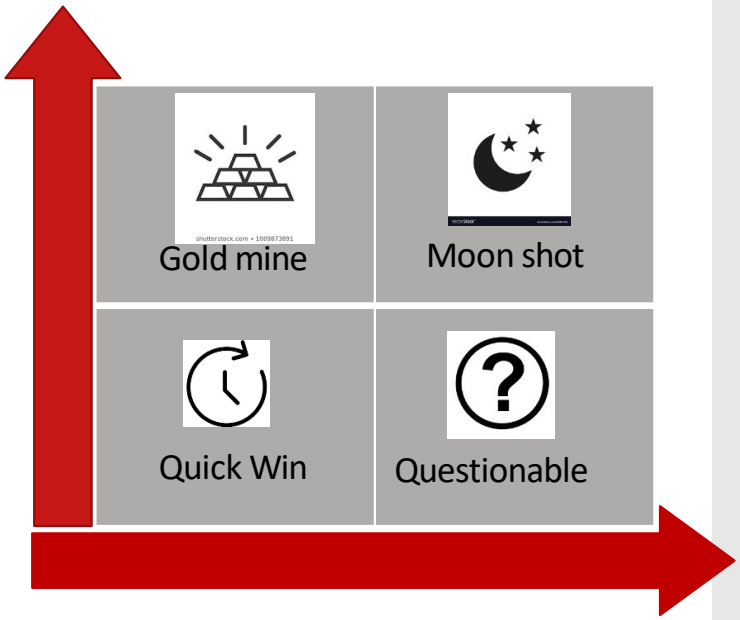
Estimate and compare the attractiveness of different options in an unbiased manner





Gruber, M. & Tal, S. (2017) *Where to Play: 3 steps for discovering your most valuable market opportunities.* Harlow, United Kingdon: Pearson.

MARKET OPPORTUNITY NAVIGATOR



WORKSHEET 2

EVALUATE MARKET OPPORTUNITY ATTRACTIVENESS

NAME

DATE

Use this worksheet for every market opportunity you would like to evaluate.

Market Opportunity:

POTENTIAL

LOW | MID | HIGH | SUPER HIGH

COMPELLING REASON TO BUY
Unmet need
Effective solution
Better than current solutions

LOW | MID | HIGH | SUPER HIGH

MARKET VOLUME
Current market size
Expected growth

LOW | MID | HIGH | SUPER HIGH

ECONOMIC VIABILITY
Margins (value vs. cost)
Customers' ability to pay
Customer stickiness

CHALLENGE

LOW | MID | HIGH | SUPER HIGH

IMPLEMENTATION OBSTACLES
Product development difficulties
Sales and distribution difficulties
Funding challenges

LOW | MID | HIGH | SUPER HIGH

TIME TO REVENUE
Development time
Time between product and market readiness
Length of sale cycle

LOW | MID | HIGH | SUPER HIGH

EXTERNAL RISKS
Competitive threat
3rd party dependencies
Barriers to adoption

OVERALL POTENTIAL

LOW | MID | HIGH | SUPER HIGH

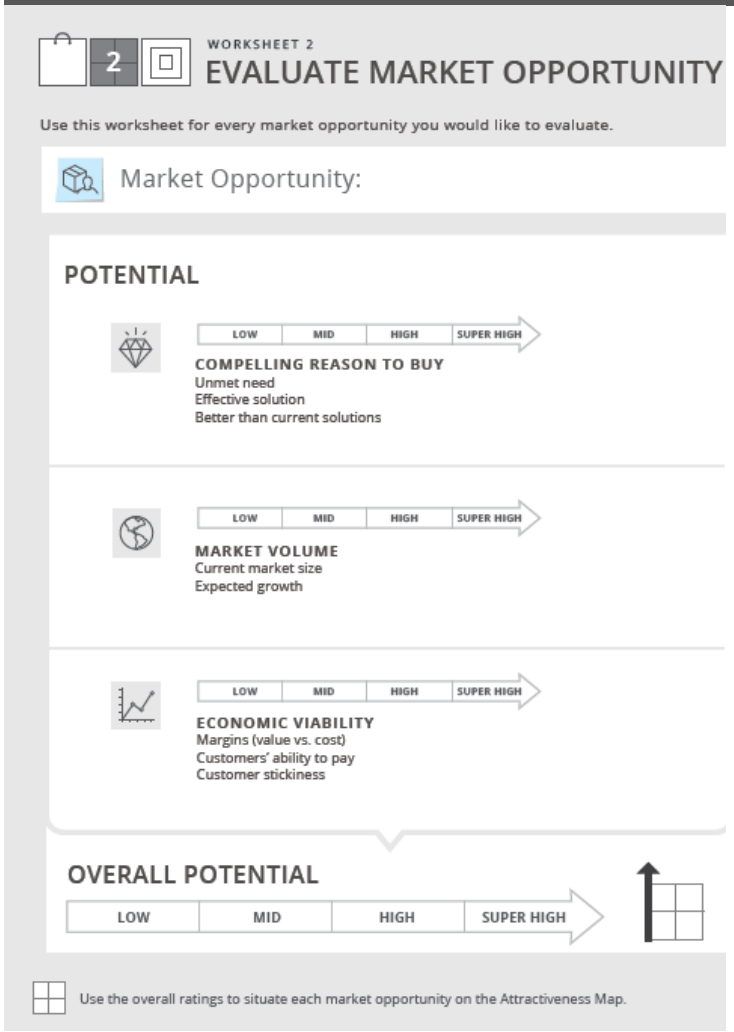
OVERALL CHALLENGE

LOW | MID | HIGH | SUPER HIGH

Use the overall ratings to situate each market opportunity on the Attractiveness Map.

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MARKET OPPORTUNITY NAVIGATOR



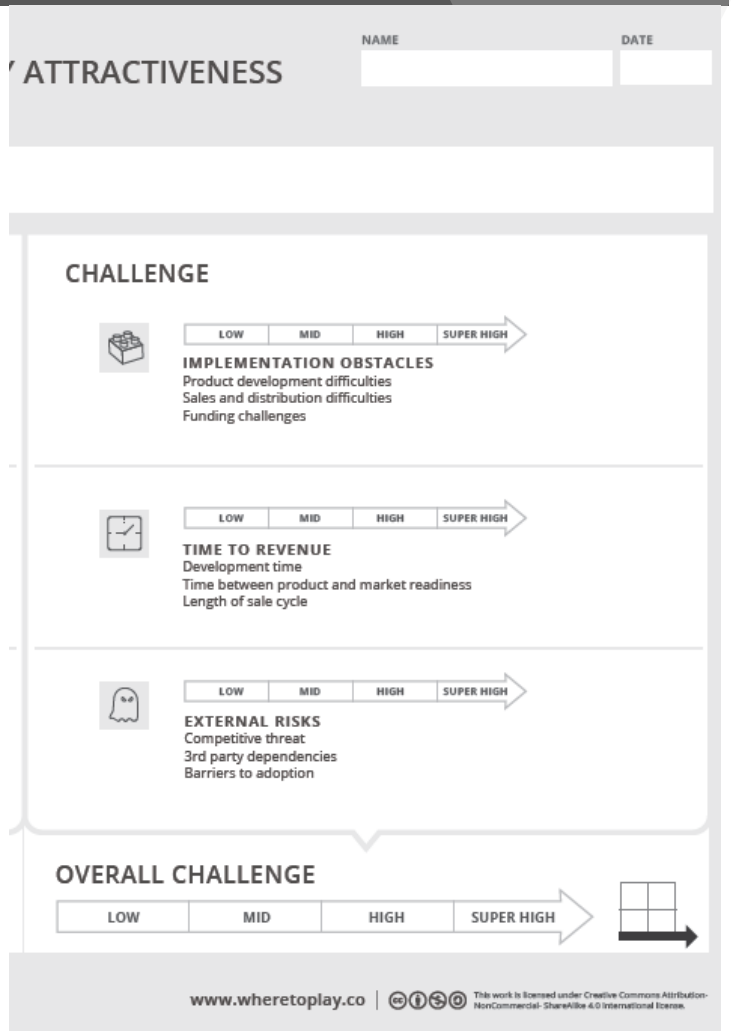
Assessing the market potential:

- How valuable is it to pursue this option?
- The opportunity it itself (without considering the capabilities you have)
- Considering the following three factors:
  - **Compelling reason to buy (without this a no-go!)**
  - **Market volume**
  - **Economic viability**

MARKET OPPORTUNITY NAVIGATOR

Assessing the overall challenge:

- How difficult it is to achieve in this market?
- In-depth examiniation of the capabilities you have at your disposal to succeed in this market
- Considering the following three factors:
  - **Implementation obstacles**
  - **Time to revenue**
  - **External risks**



MARKET OPPORTUNITY NAVIGATOR

How to use the tool?

- Together the factors provide a comprehensive overview of the key elements that determine the attractiveness of a market opportunity
- The template can be used for a first, quick assessment
- Yet, a more thorough examination is needed after a first selection. Combine qualitative and quantitative insights to thoroughly assess the opportunities resulting from your first selection (remember the data detective and journalist role of the first workshop).

CONCEPT TESTING IN ACTION



15 minutes

00 : 15 : 00

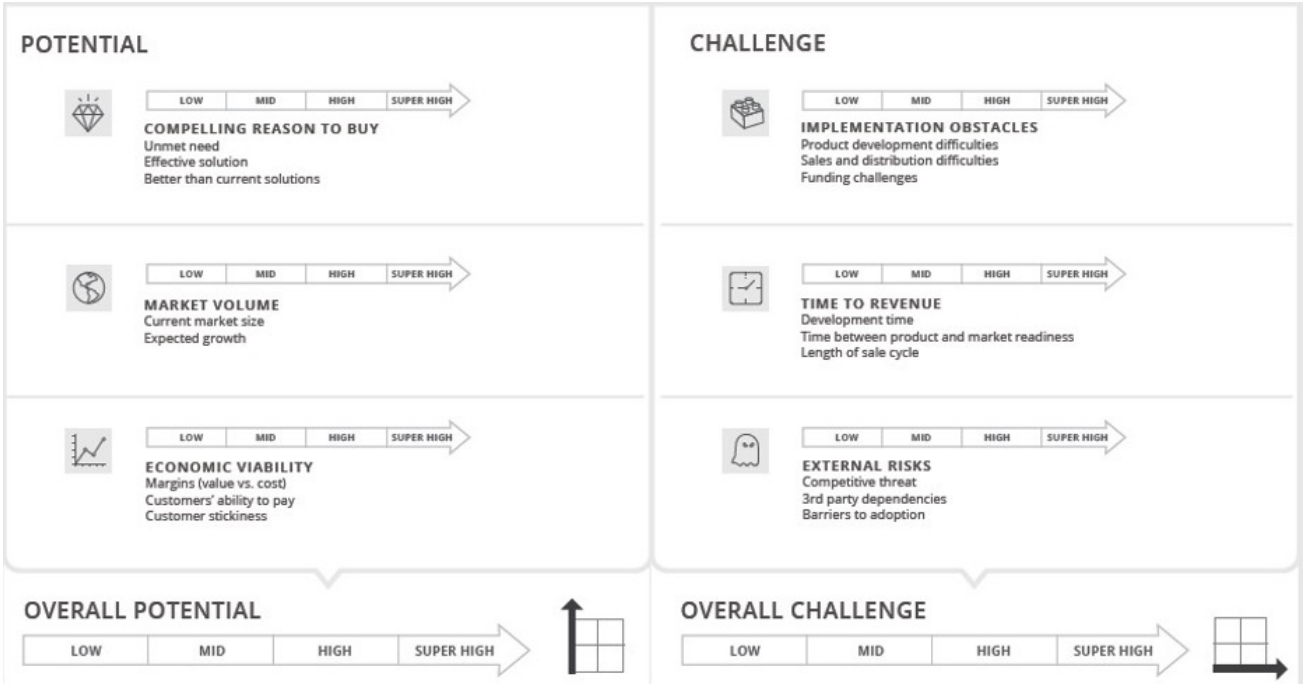
Change Clock Type

Digital

Duration:

Apply the market opportunity navigator:

- Consider which criteria are relevant, given the nature of your concept (irrelevant factors can be left out of the equation)
- Search for relevant information, if needed (”data detective” from workshop 1)



DELIVERABLES AFTER THIS WORKSHOP

Assignment 2 (see CANVAS for more information)

- Upload a slide deck where you present your concept solutions of the 3 selected ideas resulting from the previous workshops (e.g. mockups, story boards, pictures of physical prototype)

- Summarize per concept the results from both your concept testing (qualitative and quantitative; minimum of 5 participants per idea) and the application of the market opportunity navigator

- Conclude with which idea you want to move forward

- Discuss for the finally selected idea ethical concerns and how to potentially address them (see module 3 in CANVAS)